

TUGAS 04

IMPLEMENTASI UNIT TESTING DENGAN GITHUB ACTIONS

Matakuliah Implementasi Pengujian Perangkat Lunak - Pertemuan 4

Dari Konsep White-box ke Pipeline CI/CD Otomatis

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Tanggal	9 Oktober 2026
Repository	https://github.com/lutfianafaisal28-hash/Tugas04_GitHubActions

Bagian A: Git & GitHub Dasar (25 poin)

Soal 1 (10 poin)

1. Centralized VCS vs Distributed VCS. Centralized VCS menyimpan seluruh versi/history pada satu server pusat; client hanya memiliki working copy dan harus terhubung ke server untuk commit maupun melihat history. Contoh: SVN (Apache Subversion) dan CVS. Distributed VCS memberi setiap developer salinan lengkap repository beserta history-nya, sehingga commit, branch, dan merge bisa dilakukan offline lalu disinkronkan via push/pull; tidak ada single point of failure dan branching jauh lebih ringan. Contoh: Git dan Mercurial.

2. Lima Git command wajib: (1) git clone <url> - menyalin remote repository ke lokal; (2) git add . - memasukkan file ke staging area; (3) git commit -m "pesan" - menyimpan snapshot staging ke history lokal; (4) git push origin main - mengirim commit lokal ke remote; (5) git pull origin main - mengambil dan menggabungkan perubahan remote ke lokal. Pendukung: git branch, git checkout, git merge, git log --oneline, git status.

3. Fungsi .gitignore. File teks berisi pola path yang harus diabaikan Git agar tidak di-track/commit, misalnya hasil build (target/), file IDE (.idea/, .vscode/, *.iml), file OS (.DS_Store, Thumbs.db), dan file coverage (*.exec).

Soal 2 (15 poin) - Praktik Git (telah dieksekusi, history ada di repository)

```
git clone https://github.com/lutfianafaisal28-hash/Tugas04_GitHubActions.git
cd Tugas04_GitHubActions
# isi README.md dengan nama + NIM
git add README.md .gitignore
git commit -m "docs: add README"
git push -u origin main
git branch feature-test
git checkout feature-test
echo "file test untuk praktik branch feature-test" > test.txt
git add test.txt
git commit -m "test: add test file"
git push -u origin feature-test
git checkout main
git merge feature-test
git push origin main
```

History commit: docs: add README > test: add test file (branch feature-test, di-merge ke main) > feat: implement unit testing > docs: add jawaban > docs: add laporan PDF.

Bagian B: Unit Testing Framework (35 poin)

Soal 3 (15 poin) - validasiPendaftaran, 5 test case independent path

#	Input (nama, umur, email, setuju)	Expected
P1	"" , 20, user@mail.com, true (dan null)	Nama harus diisi
P2	"Budi", 15, budi@mail.com, true	Umur minimal 17 tahun
P3	"Budi", 20, budimail.com, true	Email tidak valid
P5	"Budi", 20, budi@mail.com, false	Harus menyetujui syarat dan ketentuan
P-ok	"Budi", 20, budi@mail.com, true	Pendaftaran berhasil

Implementasi JUnit: Tugas/Tugas04/src/test/java/com/stmik/PendaftaranValidatorTest.java (6 method @Test termasuk kasus nama null). Pola Arrange-Act-Assert; setup objek via @BeforeEach.

Soal 4 (20 poin) - StudentGrade

1. Cyclomatic Complexity calculateGrade(). Node predikat: $\text{score} < 0 \parallel \text{score} > 100$ (1 if + 1 operator $\parallel$ = 2), $\text{score} \geq 80$, ≥ 70 , ≥ 60 , ≥ 50 (4) sehingga total keputusan = 6, $V(G) = 6 + 1 = 7$. (Catatan: materi slide menghitung $\parallel$ sebagai satu sehingga menyebut 6; perhitungan ketat dengan operator logika memberi 7. Jumlah test case minimal mengikuti $V(G)$.)
2. Independent Path: (1) skor invalid -> throw IllegalArgumentException; (2) ≥ 80 -> A; (3) 70-79 -> B; (4) 60-69 -> C; (5) 50-59 -> D; (6) 0-49 -> E.
3. JUnit: StudentGradeTest.java - 7 test calculateGrade (2 invalid + A/B/C/D/E) + 4 test isPassed (lulus 70, batas 60, tidak lulus 50, invalid -1/120). Verifikasi: assertEquals, assertTrue/False, assertThrows.

Bagian C: GitHub Actions (25 poin)

Soal 5 (15 poin) - Analisis workflow Python CI

1. Nama workflow: Python CI.
2. Trigger: push ke branch main dan develop, serta pull_request ke main.
3. Fungsi strategy matrix (python-version: ['3.10','3.11','3.12']): menjalankan job test paralel tiga kali, masing-masing satu versi Python, untuk memastikan kompatibilitas lintas versi.
4. Steps: perlu dikoreksi - YAML yang diberikan hanya memuat 4 steps, bukan 5: (1) Checkout code (actions/checkout@v4), (2) Setup Python sesuai matrix (actions/setup-python@v5), (3) Install dependencies (pip install pytest pytest-cov), (4) Run tests (pytest --cov=src --cov-report=xml). Ada ketidaksesuaian antara soal (meminta 5) dan isi YAML (4).

Soal 6 (10 poin) - Workflow Java (file .github/workflows/java-test.yml)

```
name: Java Unit Test
on:
  push:
    branches: [main, develop]
  pull_request:
    branches: [main]
jobs:
  test:
    runs-on: ubuntu-latest
    defaults:
      run:
        working-directory: Tugas/Tugas04
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - name: Setup JDK 17
        uses: actions/setup-java@v4
        with:
          java-version: '17'
          distribution: 'temurin'
          cache: maven
      - name: Run tests dengan mvn test
        run: mvn -B test
      - name: Generate coverage report dengan mvn jacoco:report
        run: mvn -B jacoco:report
      - name: Upload artifact dengan nama "jacoco-report"
```

```
uses: actions/upload-artifact@v4
with:
  name: jacoco-report
  path: Tugas/Tugas04/target/site/jacoco/
```

Catatan: working-directory dipakai karena project Maven berada di Tugas/Tugas04 sesuai format pengumpulan.

Bagian D: Code Coverage & Integrasi (15 poin)

Soal 7 - Analisis coverage report

1. Total statement = $10 + 15 + 20 + 5 = 50$. 2. Total miss = $0 + 2 + 8 + 1 = 11$. 3. Coverage terendah: src/payment.py, 60%. 4. Untuk payment.py: buka laporan HTML, identifikasi 8 baris/cabang yang miss, tambah test untuk branch if/else, exception, dan edge case pembayaran yang belum tersentuh, lalu jalankan ulang pytest --cov. 5. Coverage 78% belum cukup baik - target materi 80-90% line coverage; 78% di bawah ambang dan payment.py hanya 60% sehingga risiko bug tinggi. Perlu ditingkatkan minimal $\geq 80\%$ (ideal $\geq 90\%$ untuk modul kritis).

Hasil JaCoCo project ini (bukti Gambar 5): 100% instruction, 97% branch, 44/44 lines, 10/10 methods - di atas target.

Bagian E: Studi Kasus Terpadu - Bonus (10 poin)

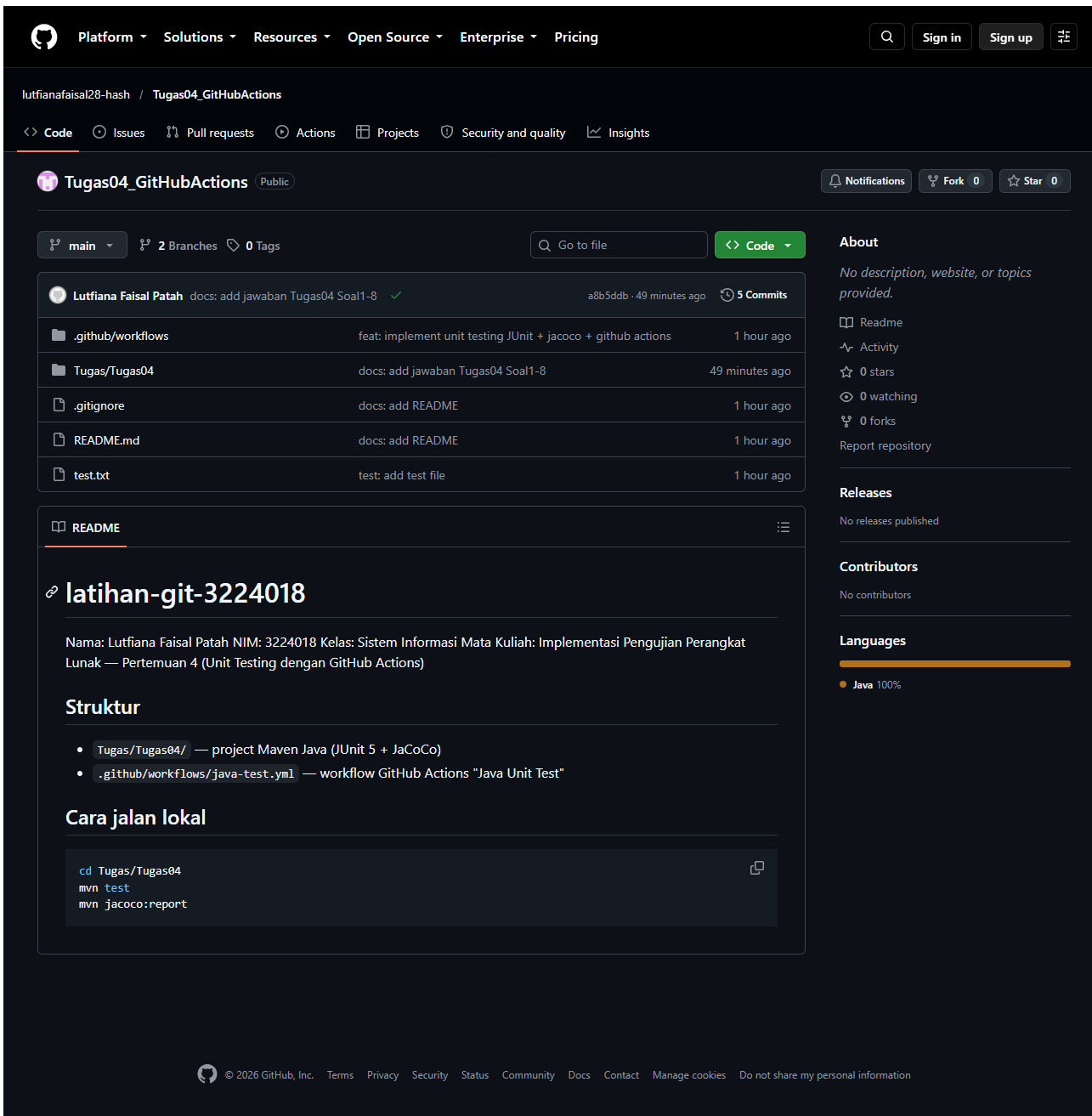
Soal 8 - BankAccount

Kelas com.stmik.BankAccount: deposit(amount), withdraw(amount), getBalance(); validasi: nominal harus positif, withdraw tidak boleh melebihi saldo, saldo awal tidak boleh negatif. Unit test (BankAccountTest.java) 9 test case: saldo awal nol, deposit berhasil, deposit negatif/nol ditolak, withdraw berhasil, withdraw melebihi saldo ditolak, withdraw negatif ditolak, konstruktor negatif ditolak, konstruktor positif berhasil. Workflow dan coverage memakai pipeline yang sama; hasil lokal 26/26 test hijau.

Link Repository GitHub

https://github.com/lutfianafaisal28-hash/Tugas04_GitHubActions

Lampiran - Bukti Screenshot



Gambar 1 - Repository GitHub.

Gambar 2 - Folder Tugas/Tugas04 di GitHub.

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Files

main

Go to file

.github

Tugas/Tugas04

src

JAWABAN_TUGAS04.md

pom.xml

.gitignore

README.md

test.txt

Tugas04_GitHubActions / Tugas / Tugas04

Lutfiana Faisal Patah docs: add jawaban Tugas04 Soal1-8 ✓ a8b5ddb · 50 minutes ago History

Name	Last commit message	Last commit date
..		
src	test: cover BankAccount initialBalance positif (26 tests green)	52 minutes ago
JAWABAN_TUGAS04.md	docs: add jawaban Tugas04 Soal1-8	50 minutes ago
pom.xml	feat: implement unit testing JUnit + jacoco + github actions	1 hour ago

Gambar 3 - Pipeline GitHub Actions SUCCESS + artifact jacoco-report.

```
Git Bash - mvn test - Tugas/Tugas04

$ cd Tugas/Tugas04
$ mvn -B test
[INFO] Building Tugas04 GitHub Actions - 3224018 Lutfiana Faisal Patah 1.0-SNAPSHOT
[INFO] Running com.stmik.BankAccountTest
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0 -- in com.stmik.BankAccountTest
[INFO] Running com.stmik.PendaftaranValidatorTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0 -- in com.stmik.PendaftaranValidatorTest
[INFO] Running com.stmik.StudentGradeTest
[INFO] Tests run: 11, Failures: 0, Errors: 0, Skipped: 0 -- in com.stmik.StudentGradeTest
[INFO] Results:
[INFO]
[INFO] Tests run: 26, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
```

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Java Unit Test

docs: add jawaban Tugas04 Soal1-8 #1

Sign in to view logs

Triggered via push 8 minutes ago

lutfianafaisal28-hash pushed · a8b5ddb

main

Status

Success

Total duration

23s

Artifacts

1

java-test.yml

on: push

test

18s

Annotations

2 warnings and 1 notice

test

Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/checkout@v...

Show more

test

setup-java v4 is deprecated and will no longer receive updates. Please migrate to actions/setup-java@v5.

test

"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/act...

Show more

Artifacts

Produced during runtime

Name	Size	Digest
jacoco-report	46.2 KB	sha256:bd720eef981634aef956d2892e6191...

Gambar 4 - Hasil mvn test lokal (BUILD SUCCESS, 26 tests).

Gambar 5 - Coverage report JaCoCo lokal.

Laporan ini disusun berdasarkan implementasi yang dijalankan dan terverifikasi pada 9 Oktober 2026.

Tugas04 GitHub Actions - 3224018 Lutfiana Faisal Patah

Element	Missed Instructions	Cov.	Missed Branches	Cov.	Missed	Cxty	Missed	Lines	Missed	Methods	Missed	Classes
 com.stmik	100%	100%	97%	97%	1	29	0	44	0	10	0	3
Total	0 of 147	100%	1 of 38	97%	1	29	0	44	0	10	0	3